

Problem Statement of Bank Loan Risk Analysis Project -

Financial institutions face a critical challenge in accurately assessing the credit risk of loan applicants while maintaining portfolio growth and minimizing defaults. Traditional risk evaluation methods often rely on isolated factors such as income or credit history, which can lead to incomplete or biased lending decisions. With the increasing availability of customer, credit, and behavioral data, there is a strong need for a consolidated, data-driven approach that provides a holistic view of customer risk.

The objective of this project is to perform an **end-to-end credit risk and customer behavior analysis** using structured loan, credit, and payment data. The project aims to transform raw, multi-table financial data into meaningful customer-level insights that help identify high-risk and low-risk borrower segments. By integrating customer demographics, previous loan applications, installment payment behavior, credit card utilization, and bureau credit summaries, the analysis seeks to uncover patterns that influence loan default.

The solution focuses on cleaning and transforming large datasets, aggregating transactional data to the customer level, and building an interactive analytical dashboard. Key business questions addressed include:

- Which customer segments are more likely to default?
- How do income, age, education, and employment status impact credit risk?
- What is the relationship between past loan behavior, payment discipline, credit utilization, and default probability?
- Which customer profiles represent high exposure with low risk, and which should be avoided for lending?

The final outcome is a **Power BI dashboard supported by SQL analysis** that enables stakeholders to monitor portfolio health, evaluate risk drivers, and support data-driven lending decisions. This project demonstrates how structured data analysis and visualization can improve credit risk assessment, enhance decision-making transparency, and support smarter financial strategies.